From public endpoints to comparable doses

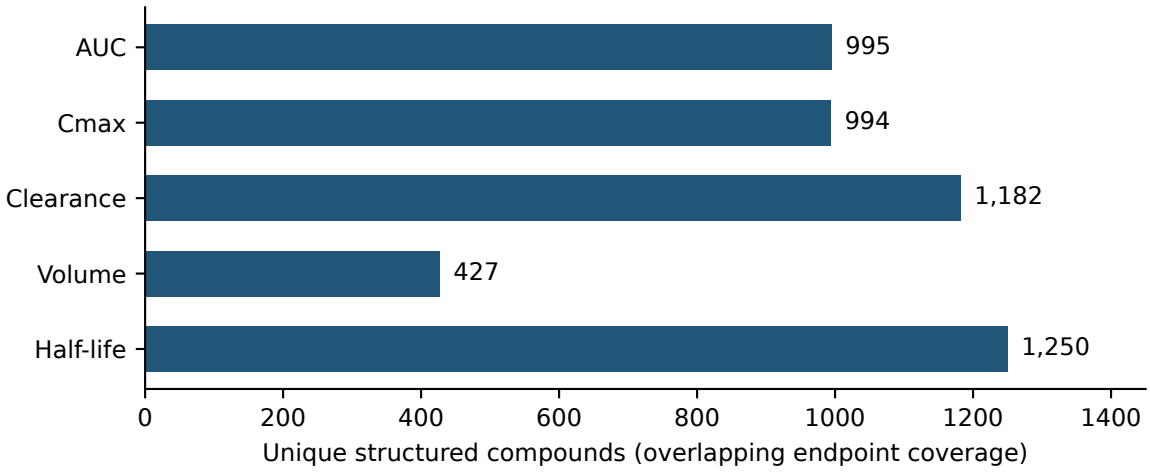
When can initial PK measurements predict exposure at an unobserved dose?

Research motivation

A useful cross-dose benchmark needs more than molecular coverage: it needs matched experimental conditions, explicit dose levels, and a genuinely held-out exposure. This study tests how far simple empirical scaling can go.

Dataset A • molecular PK coverage

2,713 structure-resolved compounds • 6,459 eligible endpoint records
5,133 excluded records retained for audit; 355 compounds have both AUC and Cmax.



Dataset B • strict cross-dose benchmark

59 comparable series • 21 compounds • 19 independent source studies
42 AUC-eligible series • 33 Cmax-eligible series • 16 eligible for both
19 compounds have at least three comparable doses; 19 have resolved structures.

Provenance and comparability

A: ChEMBL 37 rat ADME tranche, measured-animal PKSmart Table S2, and structured B endpoints. B: CEBS (47 series) and reviewed publications (12 series). Source IDs, hashes, original units and QC decisions are preserved. Route, matrix, AUC window and protocol differences remain separate.

The distinction matters

Dataset A is not cross-dose validated. Its 627 publication/study proxies are not verified independent experiments. Dataset B supports 513 episodes, but only 19 linked compound–study evaluation units overall.

Sources: ChEMBL (CC BY-SA 3.0); PKSmart, doi:10.1186/s13321-025-01066-5 (CC BY 4.0).
CEBS/PMC citations and article-specific terms: reports/data_inventory.md. No raw dataset is included.

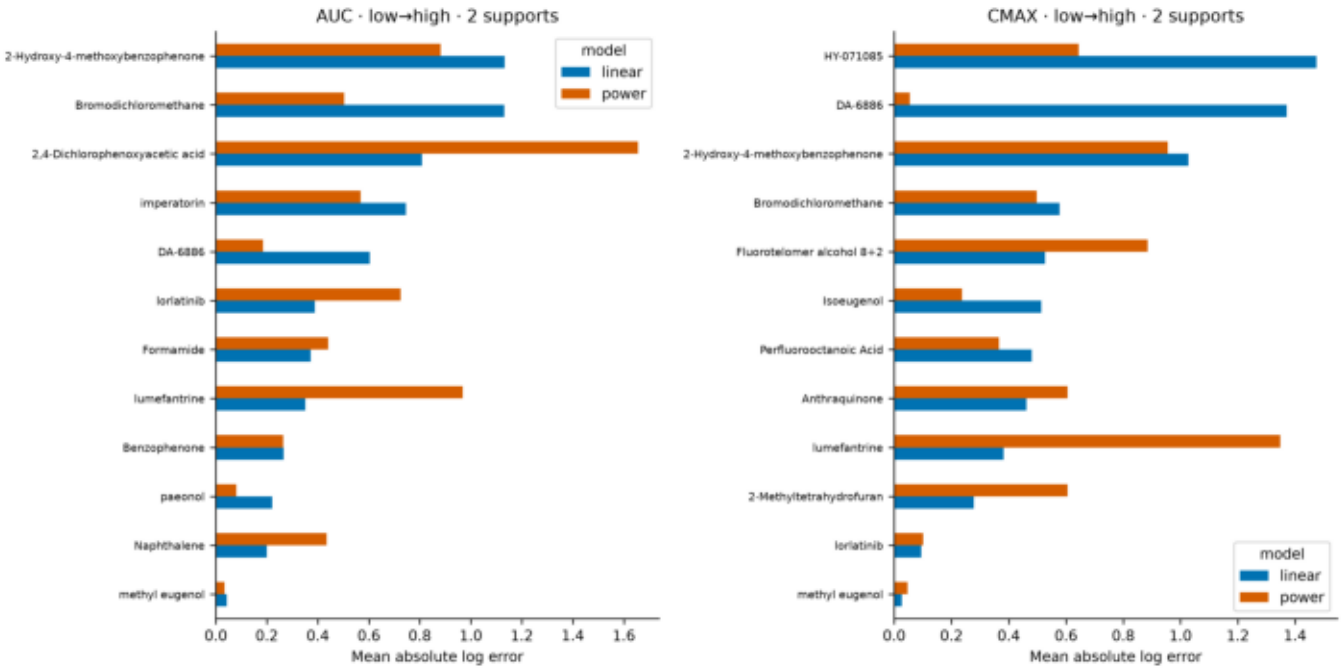
Empirical extrapolation: what the data show

Two support doses → a higher held-out dose • AUC and Cmax evaluated separately

Verified baseline comparison

Endpoint	Model	Log MAE	Log RMSE	Mean ARE	Units
AUC	Linear	0.525	0.700	38.1%	11
AUC	Power law	0.583	0.808	61.9%	11
Cmax	Linear	0.601	0.793	50.9%	12
Cmax	Power law	0.529	0.716	60.1%	12

Natural logs; ARE = absolute relative error. Equal series/component weighting.
Only 11 independent AUC units and 12 Cmax units; repeated episodes are not replicates.



Compound-level errors vary substantially. Existing figure 06; selected, correlated strata.
Methyl eugenol is favorable; bromodichloromethane AUC and HY-071085 Cmax are challenging.

Interpretation and limits

Power-minus-linear log-MAE differences: AUC +0.058 (95% exploratory interval −0.176 to +0.308); Cmax −0.072 (−0.391 to +0.231). Both include zero. Power law does not consistently outperform linear scaling; nonlinear trends do not identify a mechanism.

Next steps

Acquire more independent, comparable dose series and animal-level uncertainty before evaluating molecular or mechanistic hybrid models. Complex molecular ML was not evaluated; robust scaffold tests and calibrated prediction intervals remain unsupported.

Figure data: CEBS and source publications listed in data_inventory.md; includes lorlatinib
(PMC12570999, CC BY-NC 4.0). Retain source-specific terms; figure is not covered by the code MIT license.